

IST DBM 2025 PROJECT

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DOMAIN

Domain Overview

Our project models a restaurant ecosystem across multiple cities.

The database stores information about cities, restaurants, employees, customers, food items, and food orders.

Purpose of the System

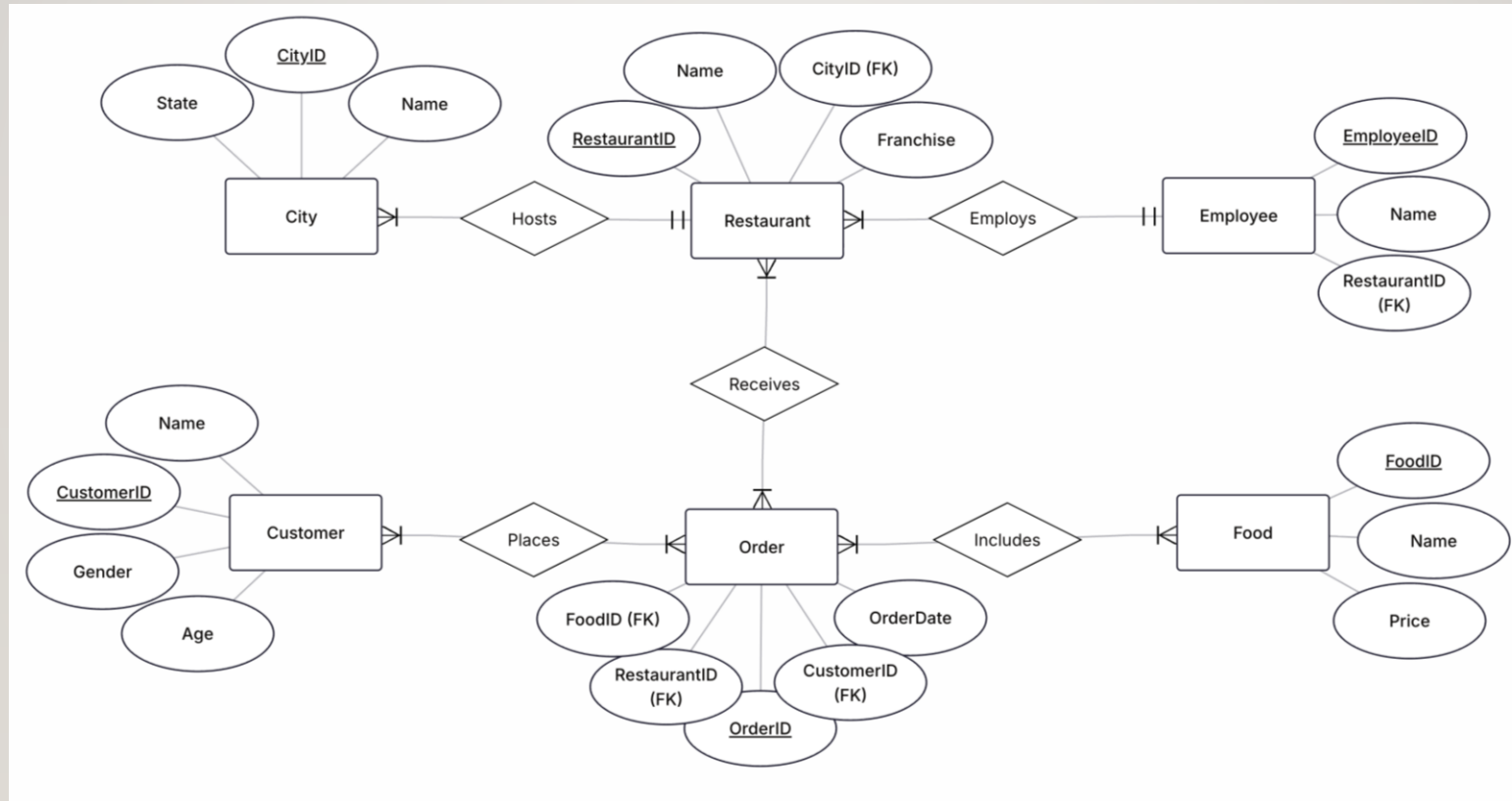
- Track restaurant locations and franchises
 - Store customer orders and food items
- Analyze customer behavior and restaurant performance
- Support advanced queries about popularity, demographics, and sales



NATURAL LANGUAGE QUERIES

1. What is the favorite dish for males?
2. Which city has the most seniors eating?
3. What's the most popular franchise in Louisiana, LA?
4. For every customer, find a restaurant in which they spend the most money in one day.
5. Find the restaurants, in which customers and employees have a different favorite dish.
6. List the restaurants, which had the most customers per month last year.

ER DIAGRAM



ENTITIES EXPLAINED

City

- **Attributes:** cityID (PK), name, state
- **Description:** Represents the geographic location. Cities host multiple restaurants.
- **Relationship:** City — hosts → Restaurant (1-to-Many)

Restaurant

- **Attributes:** restaurantID (PK), name, franchise, CityID (FK)
- **Description:** Represents an individual restaurant location, part of a franchise.
- **Relationships:** Restaurant — employs → Employee (1-to-Many)

Restaurant — receives → Order (1-to-Many)

Employee

- **Attributes:** employeeID (PK), name, restaurantID (FK)
- **Description:** Employees working at a specific restaurant.
- **Relationship:** Belongs to exactly **one** restaurant.

Customer

- **Attributes:** customerID (PK), name, age, gender
- **Description:** Customer who can place orders.
- **Relationship:** Customer — places → Order (1-to-Many)

Order

- **Attributes:** orderID (PK), orderDate, restaurantID (FK), customerID (FK), foodID (FK)
- **Description:** A purchase made at a restaurant by a customer.
- **Relationships:** Order — includes → Food (Many-to-Many through an inclusion relationship)

Food

- **Attributes:** foodID (PK), name, price
- **Description:** Menu items offered across restaurants.
- **Relationship:** Linked to orders via a Many-to-Many “includes” relationship.

QUERY 1:

What is the favorite dish for men?

$\pi\{f.name\} (\sigma\{c.gender='male'\} ((order \bowtie\{o.customer_id=c.id\} customer) \bowtie\{o.food_id=f.id\} food))$

```
query1 = psql.read_sql("""
    SELECT f.name AS favorite_dish, COUNT(*) AS orders_count
    FROM orders o
    JOIN customer c ON o.customer_id = c.id
    JOIN food f ON o.food_id = f.id
    WHERE c.gender = 'male'
    GROUP BY f.name
    ORDER BY orders_count DESC
    LIMIT 3;
""", con)
display(query1)
```

OUTPUT:

	favorite_dish	orders_count
0	Chicken Tikka Masala	17
1	Pad Thai	15
2	Pizza	14

QUERY 2:

What's the most popular franchise in Louisiana, LA?

$\pi\{r.\text{franchise}, o.\text{id}\} (\sigma\{c.\text{state} = 'LA'\} (((\text{order} \bowtie\{o.\text{restaurant_id} = r.\text{id}\} \text{restaurant}) \bowtie\{r.\text{city_id} = c.\text{id}\} \text{city}))))$

```
query3 = psql.read_sql("""
SELECT r.franchise, COUNT(*) AS orders_count
FROM orders o
JOIN restaurant r ON o.restaurant_id = r.id
JOIN city c ON r.city_id = c.id
WHERE c.state = 'LA'
GROUP BY r.franchise
ORDER BY orders_count DESC
LIMIT 3;
""", con)
display(query3)
```

OUTPUT:

	franchise	orders_count
0	Flamingo, chilean	12
1	Ostrich	10
2	Common duiker	9

QUERIES 3 - 6:

Which city has the most seniors eating?

$\pi\{ci.name, o.customer_id\} (\sigma\{c.age \geq 65\} ((((order \bowtie\{o.customer_id=c.id\} customer) \bowtie\{o.restaurant_id=r.id\} restaurant) \bowtie\{r.city_id=ci.id\} city)))$

For every customer, find a restaurant in which they spend the most money in one day.

$R = (order \bowtie\{o.customer_id = c.id\} customer) \bowtie\{o.food_id = f.id\} food \bowtie\{o.restaurant_id = r.id\} restaurant$

$R_{final} = \pi\{c.id, c.name, o.restaurant_id, r.name, o.order_date\}(R)$

Find the restaurants, in which customers and employees have a different favourite dish.

$CustOrders = order \bowtie\{o.restaurant_id = r.id\} restaurant - EmpFav = employee \bowtie\{e.restaurant_id = r.id\} restaurant - R_{final} = \pi\{r.id, r.name\}(R_{diff})$

$R_{diff} = \sigma\{o.food_id \neq e.favourite_food_id\}(CustOrders \bowtie\{r.id = e.restaurant_id\} EmpFav)$

List the restaurants, which had the most customers per month last year.

$R1 = \sigma\{YEAR(o.order_date) = YEAR(CURRENT_DATE)-1\}(order) \quad R_{final} = \pi\{r.name, DATE_TRUNC('month', o.order_date), o.customer_id\}(R2)$

$R2 = R1 \bowtie\{o.restaurant_id = r.id\} restaurant$

LIVE DEMO: FULL STACK IMPLEMENTATION LINK

<https://mrtunic.github.io/istdbdm25-testing-/PROJECT/data/>